

FRONT / DIRECT THE WORK

A practical prompting reference

Understand > Specify > Translate > Check > Improve

START WITH ONE KIND OF WORK

Explain first when the behavior is unclear. Translate only after agreeing on requirements. Request optimization separately from a checked baseline.

Copy these headings; fill only what matters

Heading	What to provide
Task	The specific work and its boundary: explain, decompose, translate, diagnose, or improve.
Source and context	Actual SAS source; relevant definitions; input schema; upstream dependencies; downstream expectations. Use only approved inputs.
Environment	Approved language and libraries; actual versions; available execution tools. State restrictions rather than letting the model install its preferences.
Required behavior	Missing values; ordering; grouping; types; calculations; output structure. Name any intended behavior change instead of disguising it as translation.
Deliverable	The exact output: explanation, code, mapping, or diagnosis. Specify whether this step is response-only and exclude unrelated changes.
Uncertainty and checks	Flag missing context and conflicts. State expected results and evidence requirements. Do not invent a rule or claim checks were run without an execution record.

Three useful starters

Understand: “Explain this SAS step without rewriting it. Identify inputs, outputs and dependencies. Separate source-supported behavior from assumptions.”

Translate: “Implement only the agreed transformation in Python. Preserve the stated behavior and interface. Return code without executing it.”

Diagnose: “For this input, expected is X and observed is Y. Use the attached code and output to identify the mismatch. Change only the affected behavior.”

How: State consequential choices in ordinary language.

Why: Explicit choices are easier to inspect and check.

Why not: Long prompts, role-play, and confident wording are not substitutes for requirements and evidence.

BEFORE YOU PRESS SEND

Is the input authorized? Is the task bounded? Is relevant context present? Have you excluded credentials and unrelated files? Are the expected outputs or acceptance rules clear?

Print at actual size on US Letter. Duplex: flip on the long edge. This sheet is a pilot aid, not an approved operating procedure.

[BACK / CHECK THE RESULT](#)

An evidence and repair reference

Different code may meet the same requirements. Repeated code may still be wrong.

Check	Ask yourself	MEAN example
Known answers	Did all expected fixture values match, including missing positions?	1, 2, 3 -> 2.0 4, missing, 6 -> 5.0 All missing -> missing
Discriminating cases	Would the checks catch a plausible but wrong interpretation?	Zero remains zero. A lone -6 remains -6.
Preservation	Did required inputs and structure survive?	No input mutation. Same labels, row order and index. Numeric average appended.

These are the MEAN pilot requirements and shared fixture expectations, not a record of SAS execution. For NaN, use `pandas.isna()` instead of equality. [SAS; Fixture; Missing]

Repair with evidence

Preserve the failed result. Provide the relevant source, inputs, expected and observed outputs, code, and error details. Ask for a bounded diagnosis and correction. Inspect the change, then rerun all checks.

Pause rather than guess

Seek clarification for missing definitions, conflicting requirements, unclear ordering or joins, unapproved dependencies, or changes to the analytical method. A reviewed change in expectations needs a new version.

Record the state, not just the answer

Keep model identifier, access application, visible settings, source/prompt/code versions, execution output and review decision. Mark unknown settings as unknown; Bedrock settings vary by model. [AWS]

Fresh conversation does not mean clean workspace. For notebooks, inspect cells, restart the kernel and run top to bottom. Restart does not delete files. Review stored outputs before sharing. [Jupyter]

Close the exercise

Record	Write one useful sentence
Required behavior	What needed to remain true?
Evidence	What did I run and what did it show?
Limits	What remains untested or unresolved?
Decision	Accept for the stated teaching scope, revise, or seek clarification.

DO NOT TURN A FAILED CHECK INTO A MOVING TARGET

Keep expected results separate from the candidate. Do not weaken checks just to obtain a pass. Three named tests do not prove universal equivalence or authorize production use.

Sources: [SAS](#) | [Fixture](#) | [Missing](#) | [AWS](#) | [Jupyter](#)